

A River Wave framework to unify river ecosystem concepts and predict ecological change

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Henry H. Hansen¹, Josh Erickson², Max Lindmark³✉, Luke McPhan^{4,5,6}, Paul Humphries^{6,7} & Eva Bergman¹

The River Wave Concept is a theoretical interpretation of river ecosystem processes that links flow to multiple river ecosystem concepts and contextualizes how a river functions with or without anthropogenic impacts. Here we propose a framework that can map the River Wave Concept to ecosystem concepts and wave metrics using widely available flow data from gauging station networks. The framework can produce a three-dimensional River Wave visualization, a time series of ecosystem process changes over time and wave metrics that can be related to organism patterns across space. Our case study contrasted Sweden's fragmented rivers with modelled free-flowing conditions and found that the observed fragmented rivers may favour specific fish life history strategies, that ecosystem functioning has shifted towards more baseflow-dominated flow compared with free-flowing rivers and that restoring all rivers to free-flowing conditions would result in more similar rivers nationally. Our study highlights how river theory can help to identify river ecosystem change and provide data to inform new analyses, which are critical for managing impacts on rivers.

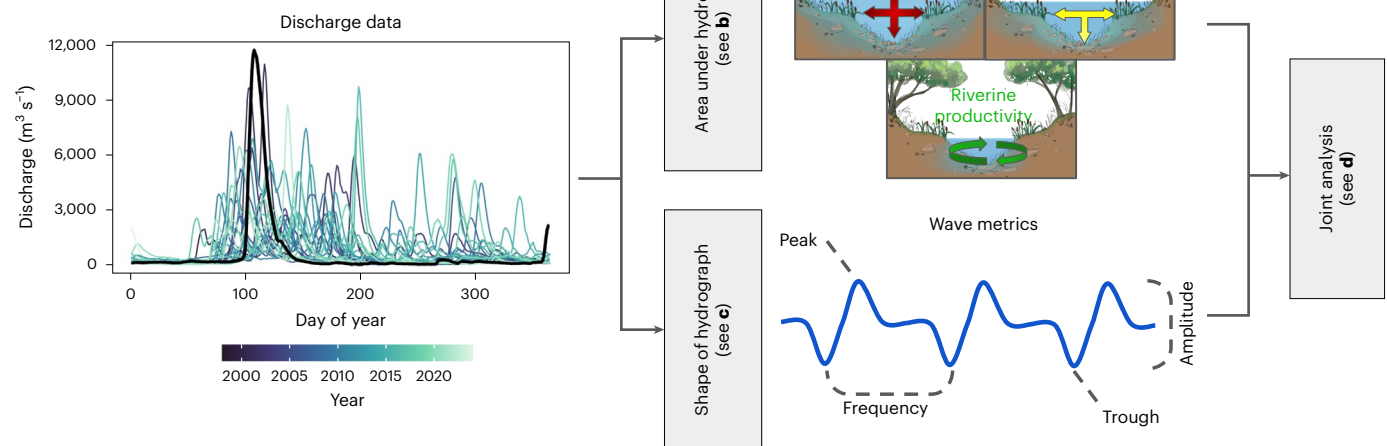
River ecosystems are experiencing an exceptional decline in biodiversity because of anthropogenic impacts and climate change^{1–3}. This decline is a direct consequence of human alterations to a river's natural flow regime⁴. Changing a river's natural flow regime results in cascading changes to a river's hydrological, geomorphological, biogeochemical and ecological processes⁵. To manage these ecosystems effectively, a variety of river ecosystem concepts (also referred to as theories or models) help to explain these processes in relation to one another. Here, three concepts are most important: (1) the river continuum concept, which focuses on the gradient of energy and resources from source to mouth of a river⁶; (2) the flood pulse concept, which emphasizes the lateral connectivity of energy and resources between the main channel and its floodplain; and (3) the riverine productivity model, which

focuses on local and in-stream production provided by local energy and resources⁷. Although these concepts have shaped our understanding of river ecology for decades⁸, they remain conceptually disjointed. Each one of them often focuses on specific types of river⁹ (for example, temperate, near-pristine and forested headwater streams) or assumes a single particular dimension of a river ecosystem (for example, longitudinal, lateral, vertical or temporal)¹⁰, limiting their transferable utility. Furthermore, it is unclear how to interpret and apply these concepts when most river flow patterns around the world are altered by dams^{11–14}.

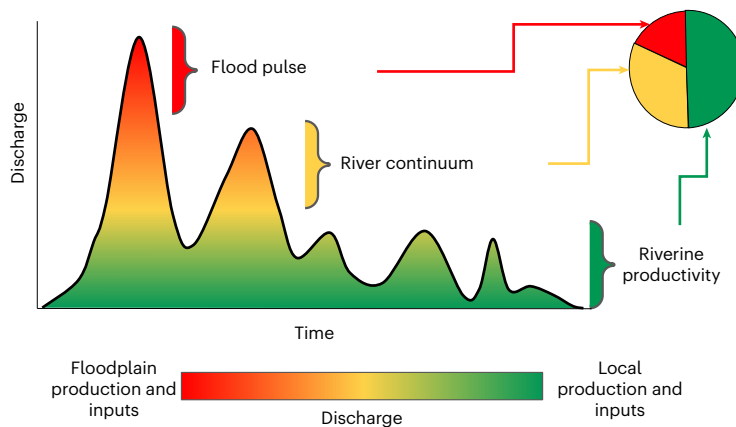
For river managers and policymakers, this conceptual misalignment creates a critical disconnect between ecological theory and management practice. Resolving this gap requires a unified conceptual framework that integrates these disparate ecosystem concepts,

¹Department of Environmental and Life Sciences, Karlstad University, Karlstad, Sweden. ²United States Forest Service Region 1, Eureka, MT, USA.

³Department of Aquatic Resources, Swedish University of Agricultural Sciences, Lysekil, Sweden. ⁴Centre for Freshwater Ecosystems, La Trobe University, Wodonga, Victoria, Australia. ⁵Environment Research Unit, Commonwealth Scientific and Industrial Research Organisation, Thurgooona, New South Wales, Australia. ⁶Gulbali Institute, Charles Sturt University, Thurgooona, New South Wales, Australia. ⁷School of Agricultural, Environmental and Veterinary Sciences, Charles Sturt University, Wagga Wagga, New South Wales, Australia. ✉e-mail: max.lindmark@slu.se

a Conceptual workflow of River Wave framework**b** Area under hydrograph

Ecological insight: proportion of ecosystem concepts over time

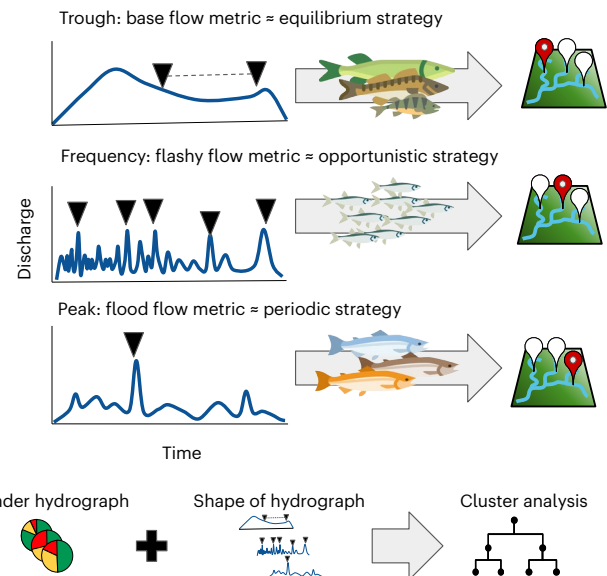
**d** Joint analysis of hydrograph area and shape

Ecological insight: cluster all stations in rivers with River Wave output

Fig. 1 | The River Wave framework uses discharge data (observed or modelled) to unify river ecosystem concepts and apply wave metrics for a joint understanding of river change. **a**, Conceptual workflow of the River Wave framework linking multi-year discharge data from a single gauge station (where individual lines represent annual hydrographs; black line denotes current year) into two parallel analytical streams: (1) delineating the area under the hydrograph to partition three foundational river ecosystem concepts—the riverine productivity model (green), the river continuum concept (yellow) and the flood pulse concept (red); and (2) relating unique features of the hydrograph's shape to wave metrics that correspond to ecologically critical flow events (for example, wave frequency measured via an annual peak flood timing metric corresponds to a specific class of organisms). The framework can be applied to multiple gauge stations to enable joint analysis of both analytical streams. **b**, Conceptual view of quantifying the area under the hydrograph. This approach unifies the three

c Shape of hydrograph

Ecological insight: life history strategy patterns over space



river ecosystem concepts into complementary, time-varying functional states defined by the interaction of discharge with the landscape (for example, a river's connectivity with the main channel, riparian zones or floodplains, implies that one of the three ecosystem concepts is a dominant concept that best describes the ecosystem processes at that particular point in time). **c**, Conceptual view of quantifying the shape of the hydrograph. This approach applies user-defined wave metrics to provide a spatial perspective of expected life history strategies. In the context of the Swedish case study, base flows select for equilibrium strategies, flashy flows favour opportunistic strategies and predictable peak floods support periodic strategies (modified from Humphries et al.^{15,37}). **d**, Joint analytical approach, which synthesizes the output of the area-under-the-hydrograph approach (**b**) with the output of the shape-of-the-hydrograph approach (**c**) using cluster analysis to provide comprehensive ecological insights for river management and policy-making.

using hydrological flow as the primary organizing variable to address a broader spectrum of contemporary river challenges. The River Wave Concept (hereafter referred to as River Wave) was proposed to unify the river continuum concept, the flood pulse concept and the riverine productivity model via flow while also incorporating anthropogenic

alterations¹⁵. By reframing river discharge as a physical wave—defined by its intensity, amplitude, wavelength and frequency—the River Wave indicates that the original river ecosystem concepts are not competing ideas but rather distinct 'functional states' that occur at different positions on the wave. The novelty of this idea is that it enables analysis of

the area under a hydrograph and the shape of a hydrograph to jointly provide a conceptual link between raw flow measurements and the river processes they help to explain. For example, as a hydrograph rises and falls over time, the area under this curve helps determine which concept is most appropriate for a river (for example, the river water level enters the floodplain, which indicates the flood pulse concept), whereas the shape of the line helps to determine the potential ecological response (for example, a peak flood event triggers fish spawning). Despite its potential to inform a broad range of river ecosystem processes, the River Wave has remained largely theoretical, due to a lack of quantitative tools to map its concepts onto real-world hydrological data. A unification of these concepts via a data-driven framework has the opportunity to translate raw hydrological data—the kind collected daily at thousands of gauging stations worldwide¹⁶—into a clear prediction of which river processes could be dominant at any given time or location.

In this Article, to demonstrate the advantages of using hydrological data to identify theory, we developed a novel quantitative framework for the River Wave. The utility of our approach enables a scalable implementation of linking empirical flow monitoring to the three river ecosystem concepts across time and space. We applied the River Wave on a nationwide scale for Swedish rivers, which have been continuously monitored by hydrological gauging stations and for which we have recently modelled free-flowing regime data for sections heavily fragmented by dams. Analysing both datasets enabled us to compare the impacts of anthropogenic barriers on perceived river ecosystem processes in relation to the River Wave. The goal of our framework was to provide a quantitative proof of concept for the River Wave, so that the variation of river processes around the world can be consistently evaluated for management decisions while also informing future research needs. To accomplish this, our objectives were: (1) to visually depict the River Wave, by creating a visualization that shows multi-dimensional patterns of how discharge links to the River Wave's hypothesized river ecosystem concepts (Fig. 1a); (2) to investigate the area under a hydrograph by analysing how river ecosystem concepts change relative to river conditions across time (Fig. 1); and (3) to investigate the River Wave metrics by relating them to organism patterns across space using the life history strategies of Winemiller and Rose¹⁷ (Fig. 1c). Both objectives (2) and (3) are analytical implementations of the River Wave, which provides a link between theorized river ecosystem concepts and observable flow data. Therefore, our final objective was to identify national-level differences between the fragmented and free-flowing datasets using the River Wave outputs from objectives (2) and (3) (Fig. 1d). As a specific example, we illustrate how we can use our framework to help managers compare the potential of fully restored flow conditions and connectivity to the observed river ecosystem processes nationally.

Theory

The proposed framework helps to fulfil the River Wave's role as a metatheory and operationalizes its central hypothesis: that the location (local or remote) and source of autochthonous production or allochthonous inputs, and the storage, transformation and transport of the material and energy derived from that production and inputs, are largely a function of the temporal or spatial position on the River Wave¹⁵. From a temporal perspective (Fig. 1b), continuous variations in discharge and associated water level alter the physical interface between a river's flow and its landscape, providing an opportunity to identify the most dominant concept among the three river ecosystem concepts. Under baseflow conditions at wave troughs, low discharge constrains energetic pathways to localized, in-stream processes where the ecosystem is dominated by autochthonous production and localized nutrient cycling, making the riverine productivity model the most applicable concept⁷. As discharge operates above baseflow but remains within bank-full capacity, longitudinal transport accelerates, shifting ecosystem processes towards the downstream delivery of

upstream-derived materials, which describes the processes associated with the river continuum concept⁶. Finally, during overbank flows at wave crests, peak discharge events overtop channel boundaries to laterally integrate the river with its adjacent floodplain, incorporating nutrient and material subsidies derived from the terrestrial–aquatic interface alongside longitudinal inputs, meaning that the flood pulse concept best describes the dominant processes¹⁸.

A spatial perspective of the River Wave (Fig. 1c) implies that the wave metrics associated with the shape of the wave, such as baseflow consistency (that is, the trough of the hydrograph indicates baseflow conditions), flashy flow (that is, amplitude) and peak flood flow (that is, frequency) events are linked to geographic location. Across an entire watershed, individual rivers may be uniquely characterized by a dominant flow event type, which in turn influences the biological response of organisms that reside there¹⁹ (for example, regular peak flood flow events predict life history strategies that rely on periodic flooding). Using discharge data on the order of multiple decades indicates that a life-history-strategy perspective may be appropriate for exploring the relationships of the River Wave to ecological communities, but the framework is not constrained to life history theory. The River Wave can accommodate a broad range of approaches (for example, community assembly) to explore the implications of specific flow events on resident biota. Finally, the potential to fully complete the unification of river ecosystem concepts is realized using data from both the shape and area of the River Wave together (Fig. 1d) to detect key differences in perceived ecosystem processes at larger macro-ecological scales⁸.

Results

River Wave visualizations

Our visualization objective involved the statistical estimation of the unique discharge breakpoints that distinguish the riverine productivity model, the river continuum concept and the flood pulse concept for each gauge station's hydrograph time series ($n = 165$). Whereas there is value in seeing every visualization at each gauge station, for brevity we show those for only three representative rivers in Sweden (Fig. 2) out of the 165 stations studied (Supplementary Information, section 1) to highlight the broadest variety of discharge patterns and their relation to river ecosystem concepts. For context, all three rivers are defined as index rivers in Sweden, which are used to track populations of returning salmon²⁰. Vindelälven (the downstream portion is monitored by the Granåker gauging station; Fig. 2a) is a northern, large, near free-flowing river that shows two characteristic attributes of free-flowing rivers: (1) the influence of a snowpack-driven water year when discharge remains very stable during winter and (2) peak flooding occurs with regular timing every year (shown as the red peaks). The Mörrumsån (the Mörrum gauging station; Fig. 2b) is a southern example that had a hydropower plant removed in 2021 in its main stem but has operating hydropower plants upstream of the gauging station. It highlights a different type of flood shape and timing compared with the Vindelälven as it shifts from conditions best described by the riverine productivity model to conditions that are more associated with the floodplain, varying in both magnitude and timing. The last example is the Rönne Å (the Forsmöllans Kraftverk gauging station; Fig. 2c), a southern example where flow regulation occurs for hydropower. The River Wave visualization of each river shows its potential interaction with the landscape and the hypothesized river ecosystem concept that best applies to that particular point in time. The timing of each concept naturally varies within and among years, which has cascading outcomes for other aspects of the ecosystem.

Changes in river ecosystem concepts across time

For our objective of mapping river ecosystem concepts across time (Fig. 1b), we averaged the daily occurrence of the riverine productivity model, the river continuum concept and the flood pulse concept for each gauge station's entire time series, where each day is described

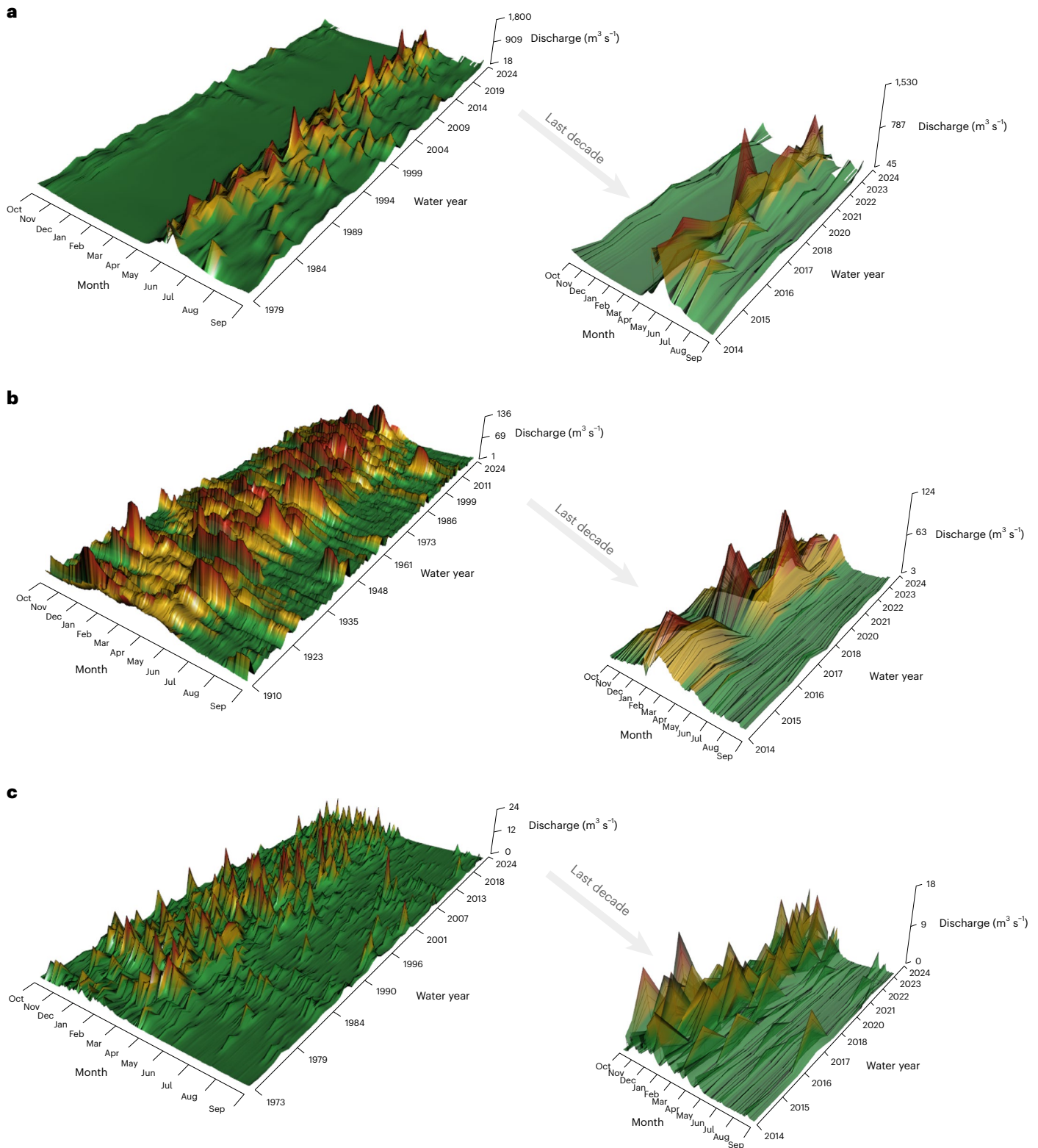


Fig. 2 | River Wave visualizations of three Swedish rivers with the observed flow regimes. Each River Wave visualization shows the discharge measured by day (for the October water year) across its time series and is colour-coded to correspond to a river ecosystem concept. Green denotes the riverine productivity model, yellow indicates the river continuum concept and red

identifies the flood pulse concept. The inset in each graph shows the final decade of the observation data. **a.** Vindelälven, a free-flowing river. **b.** Mörrumsån, a river that had a hydropower dam removed in 2021. **c.** Rönne Å, a river with hydropower operations that is no longer active (since 2018) but still has a barrier. Note the different time series for the three rivers.

only by the dominant concept. This provided a quantitative way to compare the proportions of the most relevant river ecosystem concept throughout an average year for both the observed fragmented and the modelled free-flowing datasets. Our comparison found that the riverine

productivity model was more common with the modelled free-flowing regime, as more values are depicted above the 1:1 black line in Fig. 3. This was true particularly during the warmer spring and summer months, resulting in potentially more local sources of production and nutrient

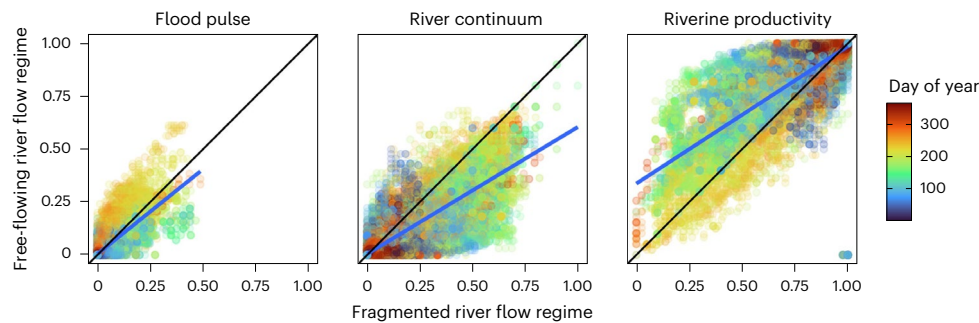


Fig. 3 | Comparison of river flow regimes under free-flowing and fragmented conditions across the three river ecosystem concepts. The regression fits (indicated by the blue lines) show the proportion of each river ecosystem concept by day comparing the modelled free-flow regime and the observed fragmented flow regimes for all of the gauge stations ($n = 165$). Each day for

each river is assigned to one of the three conceptual models. Proportions are averaged by the day of year (October water year) for each concept across all time series. The black line in each indicates the 1:1 relationship between each flow regime, thus a station sitting on the line showed equal proportions of each river ecosystem concept.

inputs. Conditions associated with both the flood pulse concept and the river continuum concept are more common under the observed fragmented flow regime, which indicates greater contributions from adjacent riparian zones and upstream contributions. The timing of the flood pulse concept and the riverine productivity model varies more than the river continuum concept, depending on the day of the year. This is most noticeable for the day-of-year range of 100–200 (green to yellow), which shows inverse relationships (that is, green on one side of 1:1 line, yellow on the other). The proportions at the end of the water year, above day of year 300 (dark red), are closest to the 1:1 line for all river ecosystem concepts, indicating similar conditions among the two datasets for that time period. Conversely, the proportions at the beginning of the water year, below day of year 100 (blues), have the greatest variation above and below the 1:1 line for each river ecosystem theory. This indicates that the river conditions among the two datasets are more random then and not necessarily biased towards a particular river ecosystem theory. The adjusted R^2 (coefficient of determination) for the regression models ranged from 0.61–0.73 (Supplementary Information, section 2).

Relations of River Wave metrics to organism patterns across space

We linked the River Wave metrics to organism patterns by assuming that the specific life history strategies of Winemiller and Rose (that is, equilibrium, periodic and opportunistic; Fig. 1c) were dependent on specific wave metrics (for example, peak flood flow is a frequency River Wave metric that periodic fish require). We applied each metric to the entire time series for all stations, for both the observed fragmented and the modelled free-flowing datasets, and used ternary plots to identify how a station may support one of the three life history strategies (Supplementary Information, section 3 and Supplementary Table 2).

The River Wave metrics associated with fish life history strategies (peak flood timing, baseflow consistency and flashiness) showed similar patterns across all stations in the ternary plot for free-flowing rivers (Fig. 4a). A typical river under the free-flowing regime will be characterized by consistent peak flood timing (between 80 and 100), moderate baseflow consistency (between 40 and 60) and similar flashiness across latitude (between 0 and 20), with a slight inverse relationship to catchment size. Hypothesized fish life history strategies for these rivers predict conditions that will be a suitable balance between periodic and equilibrium fish but will create limited possibilities for opportunistic fish to have a competitive advantage. Thus, the River Wave predicts a spatially uniform composition of the three life history strategies across the entire country for the modelled, free-flowing rivers.

By contrast, the observed fragmented flow regimes showed patterns strongly associated with latitude and catchment size (Fig. 4b). Flashiness tends to be low for higher latitudes, whereas baseflow

consistency increases for lower latitudes. Stations at higher latitudes also have a higher consistency in peak flood timing, where two groupings are visually present. Most stations are between a periodic state (top of the triangle) and an equilibrium state (lower right of the triangle), with no stations sitting directly in a corner. Smaller catchments also tend to be more flashy than larger systems. Hypothesized fish life history strategies for the observed fragmented flow regimes would be a strong latitudinal gradient with periodic fish dominating at northern latitudes, equilibrium fish dominating at southern latitudes and opportunistic fish colonizing a few smaller catchments. For both plots, no stations are present in the opportunistic state (lower left of the triangle) and only one station had a near-zero baseflow value, putting it on the left axis of the triangle. The ternary plot provides a clear spatial understanding of how management operations (for example, fisheries regulations, stocking, dam operation) can be contextualized to a specific point location under current conditions and future possibilities that a free-flowing regime may provide.

Joint analysis of the River Wave and metric views to inform holistic river management

Our final objective used the outputs from the concepts and wave metrics to analyse how rivers in Sweden may change if all were managed for free-flowing conditions at a nationwide scale in comparison with its current state of high fragmentation. This is crucial in addressing the theory-to-practice gap as Sweden continues to move forwards with a national effort to increase river connectivity. It is currently unclear how many types of river that managers should expect after free-flowing conditions return at a national scale. Alternatively, it may be vital to understand which rivers are optimal to prioritize free-flowing conditions given limited time, money and resources. To assess how rivers may change in response to flow management, we conducted a hierarchical cluster analysis on the river ecosystem concept proportions and the River Wave metrics calculated from both the free-flowing regime data and the observed fragmented flow regime data. We reasoned that river ecosystem concept proportions (per cent of each concept for the entire time series), variation in the River Wave properties (peak flood timing, baseflow consistency and flashiness) and station characteristics (catchment size) could inform possible management regimes if more rivers in Sweden were to return to a free-flowing condition. For the observed fragmented flow regime, the hierarchical cluster analysis identified three optimal clusters based on the gap statistic (gap = 0.92, standard error (SE) = 0.020; Supplementary Information, section 4 and Supplementary Fig. 4), which was validated using a k -means clustering approach that also found the ideal number of clusters to be three (Supplementary Fig. 5). For the modelled free-flowing regime data, the hierarchical cluster analysis identified two optimal clusters based on the gap statistic (gap = 0.76, SE = 0.020; Supplementary Fig. 6),

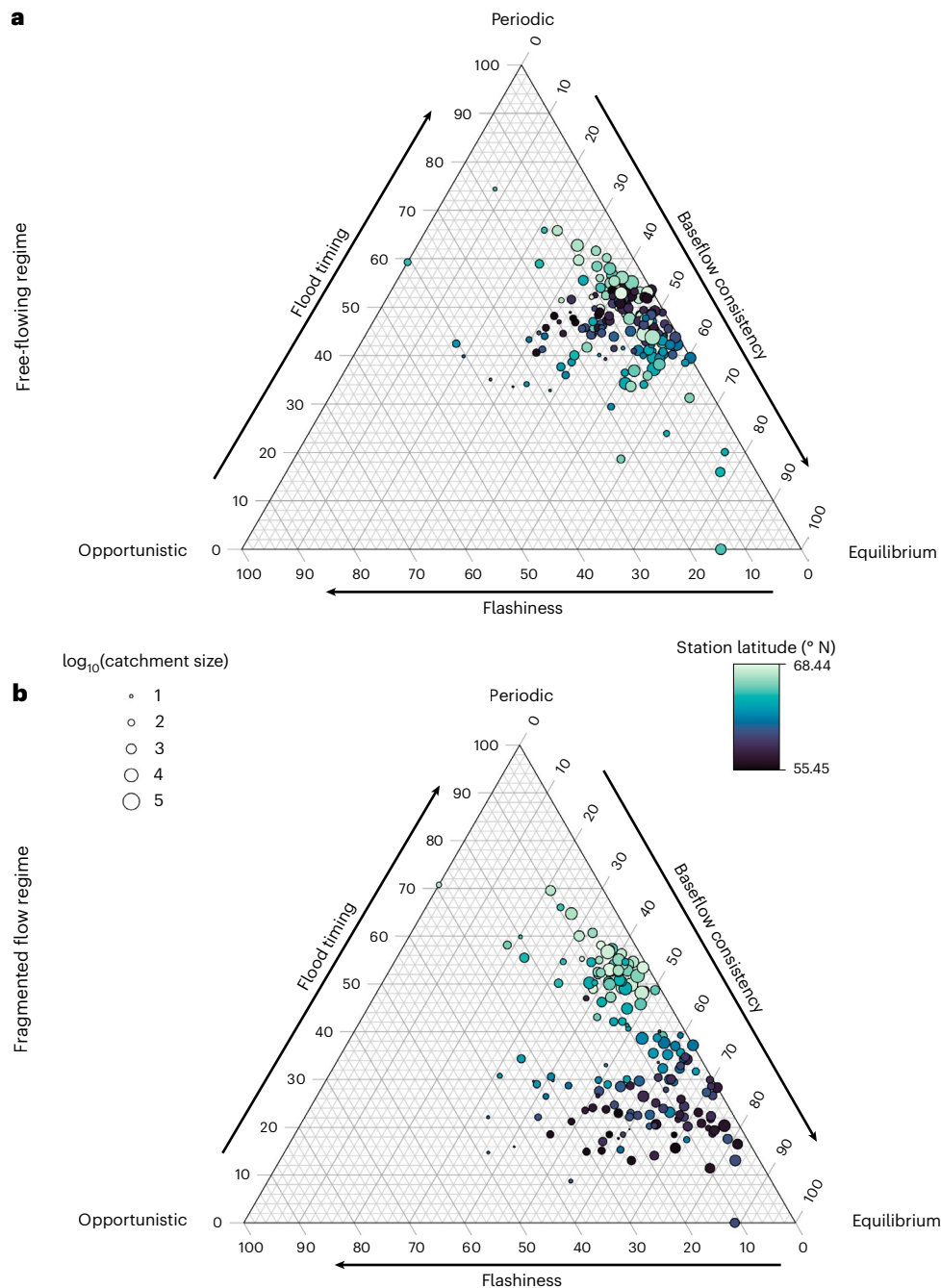


Fig. 4 | Ternary plots of expected fish life history strategy patterns.

a,b, Ternary plots of expected fish life history strategy patterns for the modelled free-flowing regime (**a**) and the observed fragmented flow regime (**b**). Data points are scaled on the basis of catchment size and are coloured by latitude. Fish life history strategies are characterized as: (1) equilibrium: high juvenile survivorship that is associated with stable flow conditions; (2) periodic: high

fecundity that is associated with regular flood events and timing; and (3) opportunistic: short generation time that is associated with highly disturbed environments. Flood timing corresponds to the coefficient of variation in annual timing of peak flow events. Baseflow consistency corresponds to the skewness of baseflow index conditions across years. Flashiness corresponds to the average volatility of flow events across years.

which was validated via *k*-means clustering that also found the ideal number of clusters to be two (Supplementary Fig. 7). Our original cluster analysis included latitude, but no agreement in cluster validation was possible so it was dropped. We found that the proportions of the individual ecosystem concepts were the strongest predictors of clusters followed by the River Wave metrics for both fragmented and free-flowing data (Supplementary Figs. 8 and 9). Most notably, we found that a free-flowing regime throughout the entirety of Sweden would have more similar rivers (that is, two clusters) than the observed fragmented flow regime (that is, three clusters) (Fig. 5). Most rivers in cluster 1 stayed in the same cluster for both datasets; by contrast,

the remaining rivers split into two different clusters for the observed fragmented flow regime data but merged together with the modelled free-flow regime data. This indicates that management should anticipate one class of rivers that do not respond dramatically to the return of free-flowing conditions and one class of rivers that will change but will result in greater similarities across the landscape.

Discussion

Our framework provides an approach for modelling river ecosystem changes across time and space by linking flow measurements to river ecosystem concepts at a national scale. The benefits of a data-driven

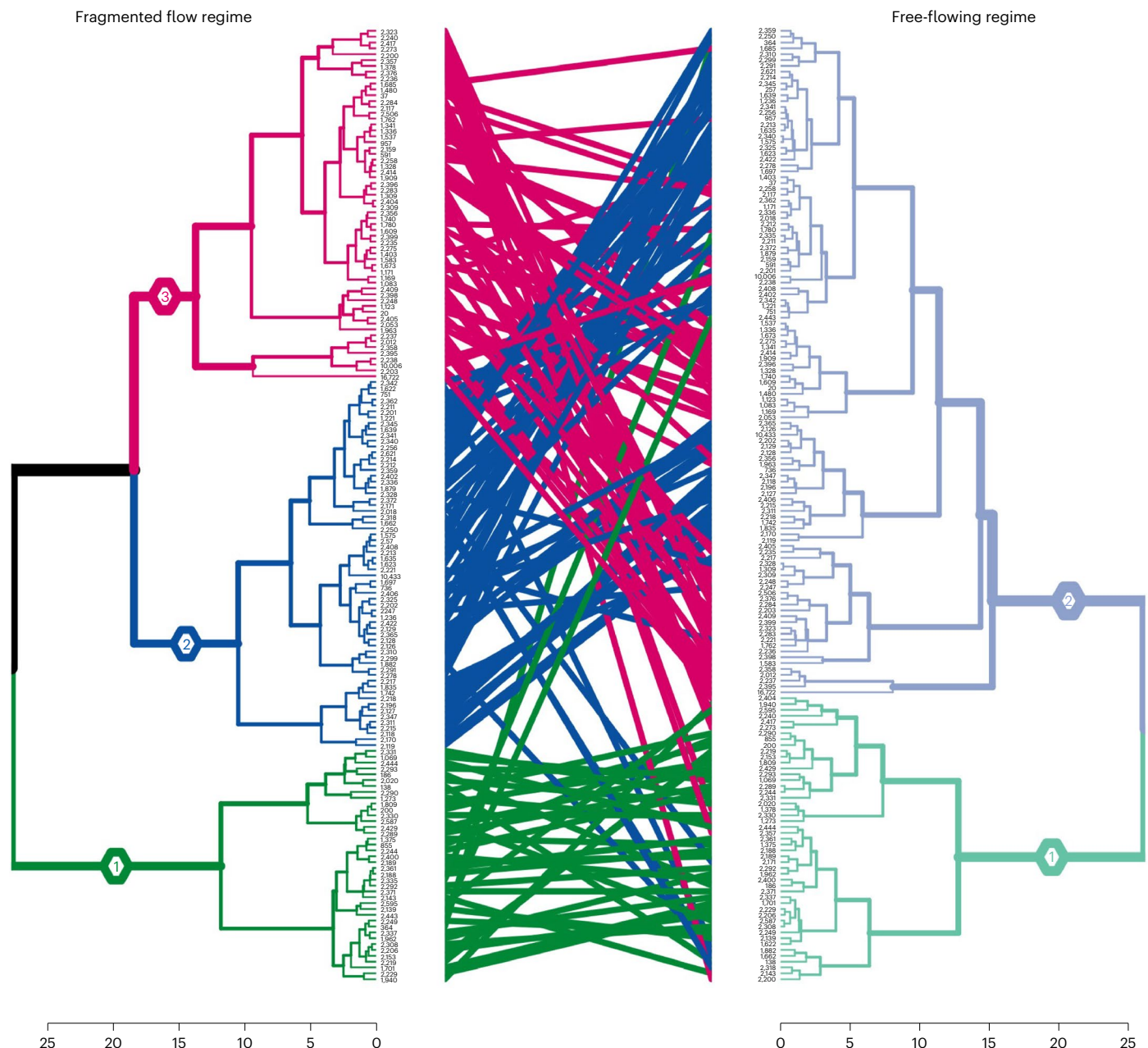


Fig. 5 | Joined dendrograms from the hierarchical cluster analysis of the observed fragmented flow regime and modelled free-flowing regime datasets using the River Wave shape and area data. The colours for each dendrogram are assigned on the basis of the clusters identified in the cluster analysis and validation. Each line connects the same station across both datasets

but is colour-coded based on the fragmented flow regime classification. The numbers along the dendrograms are the identifiers of the individual gauge stations tracked in this study. The 0–25 scale is the distance metric used for the cluster analysis. This plot shows how similar the observed fragmented rivers would be if they were managed to free-flow conditions.

view of the river are the multi-faceted insights that emerge from an integrated understanding of a river's flow via the River Wave framework. Outputs from the River Wave framework uncovered patterns of flow with direct implications for understanding river ecosystem function, fish life history strategies, ecosystem impacts from fragmentation and river management. First, the River Wave visualizations demonstrated the possible changes of ecosystem processes to a daily level of precision. Second, changes in river ecosystem concept proportions were detected between fragmented and free-flowing conditions. Third, comparison of River Wave metrics offered potential ecological community responses to ambitious restoration goals (that is, return to free-flowing conditions across the country). Last, the cluster analysis revealed new evidence of homogenizing river flow

regimes under free-flowing conditions as opposed to the observed fragmented situation.

A major advantage of the River Wave framework is its flexibility to match the unique flow characteristics of a river ecosystem. Our delineation of river ecosystem concepts, summary of proportions and application of metrics were specific to each gauge station. This is particularly important as it enables river-specific guidance for future restoration goals. For example, critical insight is needed for the European Union as its new biodiversity strategy for 2030 proposes the restoration of 25,000 km of free-flowing rivers and the recovery of ecosystem processes linked to free-flowing rivers²¹. Throughout many river systems around the globe, it is unclear which river ecosystem concept or flow metrics are most appropriate to contextualize stressors, management

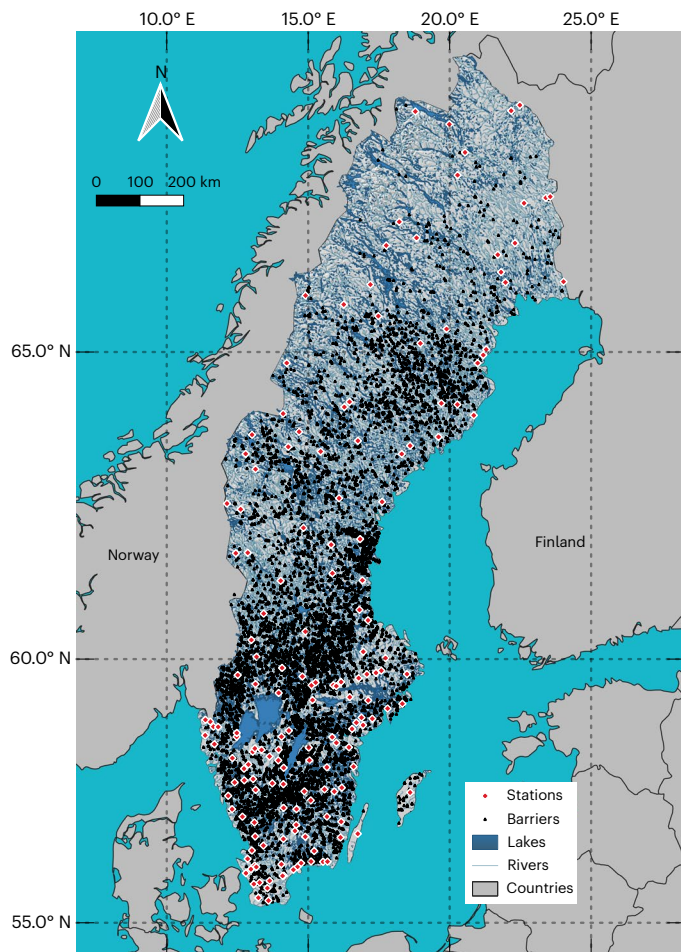


Fig. 6 | Study area map of the selected Swedish Meteorological and Hydrological Institute discharge gauging stations in relation to Sweden's landscape. Barriers are indicated with black triangles. Credit: Swedish Meteorological and Hydrological Institute under a Creative Commons license [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/).

actions and restoration expectations for each river. Our case study focused on barriers, one of the most pervasive impacts on rivers in Sweden (Fig. 6), and our analyses showed that impacts associated with barriers have shifted rivers to a point that their original free-flowing hydrologic characteristics, and associated ecosystem functions, have been drastically altered^{22,23}. Surprisingly, the cluster analysis showed that free-flowing conditions may result in more similar rivers from an ecosystem concept and wave metric perspective, which provides an important exception to the scale-dependent general opinion that free-flowing rivers are naturally more heterogeneous than fragmented ones²⁴. Barriers have the potential to change ecosystem functioning depending on the location, type of barrier and number of barriers²⁵, which presents a potential need to incorporate other river ecosystem concepts such as the serial discontinuity concept that focuses on barrier-driven ecosystem processes¹³. As long as flow information remains a key mechanism for delineating alternative or more advanced river ecosystem concepts, the River Wave can be updated and adjusted over time to reflect the uniqueness of flow alongside evolving river ecosystem concepts.

The analysis of the fragmented rivers in this study indicates large differences from the free-flowing perspective, which could be field-validated. From a fish life history perspective, we would need to relate real fish observations across time for our study area to ensure that the River Wave interpretation of flow metrics is correct. The River Wave framework provided a hypothesis that Swedish rivers are typically more conducive to supporting equilibrium species, with very

stable baseflow conditions, and that there were very few instances of extreme flashiness across time that may be needed to support other life history strategies for fish biodiversity. Our prediction of less suitable conditions for periodic fish such as salmon could possibly explain their decline in the country. The ecosystem proportions regression comparison between fragmented and free-flowing rivers predicted that the relevance of a particular concept and associated processes can change via fragmentation. Field validation of any prediction from our framework should always be considered to be a necessary follow-up to test the sensitivity of the River Wave's assumptions and the discharge data that are being used.

We found more clusters in the observed fragmented flow regime data than in those of the free-flowing regime. Sweden's granite-and-boreal-dominated landscape²⁶ probably limits large variation in river characteristics, which could explain the lower number of clusters in free-flowing regimes. The lines connecting cluster 1 fragmented to cluster 1 free-flowing (Fig. 5) also indicate that a small portion of rivers still resembles the same River Wave characteristics regardless of fragmentation. Most other rivers show that transitioning from a fragmented regime to a free-flowing regime (that is, by restoring connectivity), would result in the majority of rivers sharing similar River Wave characteristics. How such dramatic changes would influence biodiversity nationally remains an open question. Future investigations of management techniques could evaluate how different types of fragmentation may change flow patterns in a way that the River Wave can help interpret. The occurrence of more clusters during fragmentation versus free-flowing conditions indicates that some fragmented rivers still have the opportunity to recover River Wave properties that are similar to free-flowing conditions. However, the management approach to achieving this effort must recognize that some rivers may not respond well to restoration if River Wave properties are the only metric of success. This is currently of crucial importance for Sweden as it undergoes a nationwide hydropower-relicensing process that requires improved environmental measures for flow and fish habitat.

Future extensions of the framework have many possibilities. One should consider how landscape data (for example, digital elevation models) can help to statistically inform the associations of discharge-state relationships to delineate ecosystem concepts, which is central to the flood-frequency analysis that we used to delineate breakpoints. Similarly, the advantage of having modelled free-flowing data for flood-frequency analysis can be enhanced by matching the exact same time ranges of the gauge station data. A sensitivity analysis that can test the different assumptions of the River Wave and other stressors on flow regimes would help with applying the framework to a broader array of river systems. Examples such as drought and water abstraction, runoff and non-point-source pollution, among many others, would enable the framework to be more globally applicable for management and restoration, but also provide the possibility of making new theoretical advancements. In addition, one could evaluate a station's time series by water year to track the trajectory of a station to gain a better idea of system dynamics instead of an averaged snapshot, as we did in this study. Expanding the approach to include other biomes is also becoming more feasible as flow gauging stations are becoming more prevalent around the world, which in turn could assess potential river concepts at very large scales, with direct implications for local biodiversity and ecosystem functioning¹⁶.

The River Wave is still relatively new and remains to be evaluated for rivers around the globe. Our framework facilitates that task to more broadly explore its hypotheses for a variety of spatial scales, taxa and stressors. The intentional choice to unify disparate ecological concepts into a cohesive data-driven view tailored to local river ecosystem stressors pragmatically addresses the river ecosystem theory-to-practice gap. Our results from Swedish rivers revealed that fragmentation is not merely a physical alteration of flow but a potential nationwide regime shift with consequences that will affect fish

Table 1 | Overview of Winemiller and Rose's life history triangle with the original descriptions of each strategy, environmental conditions, population response to environment and relation to Sweden flow regimes

Winemiller and Rose's life history strategy	Environmental conditions	Predicted population response to environmental driver	Flow regime pattern: River Wave property measurement
Equilibrium	Increasing resource stability , competition and predation	Stable and near carrying capacity : responds to environmental traits that enable density-dependent selection	Consistency of baseflow conditions: skewness of annual baseflow index values
Periodic	Increasing magnitude and predictability in spatio-temporal variability of resources and mortality factors	Cyclical: responds to large-scale environmental variation influencing early life stage survival	Annual peak flood timing: coefficient of variation for the day of year of the annual maximum discharge event
Opportunistic	Increasing environmental disturbance, decreasing predictability in spatio-temporal variability of resources and mortality factors	Stochastic : responds to an environment that contains frequent density-independent sources of mortality	Average flashiness conditions: the Richards–Baker flashiness index value

Bold text indicates the key terms associated with flow-based drivers of the environment.

biodiversity and management priorities. Such interesting discoveries have tremendous potential to inform both theoretical and applied questions for river ecosystems. Ultimately, this framework provides a robust, scale-independent tool for predicting how ecological change propagates from daily hydrological shifts to long-term ecological changes, offering a unified path forwards for the synthesis of river ecosystem theory now and later.

Methods

Study system and flow-data collection

Sweden's rivers reside in a boreal landscape with coniferous forests in the north and mixed broadleaf forests with agriculture in the most southern part. Torneälven, Kalixälven, Piteälven and Vindelälven are Sweden's four large free-flowing rivers that are protected from hydropower development, whereas other rivers are heavily fragmented by barriers for hydropower, irrigation and municipal use (Fig. 6). The latest barrier register reports over 10,000 barriers in lotic waterways, of which approximately 2,000 barriers are hydropower dams. We downloaded the daily discharge measurements of 165 gauging stations from the Swedish Meteorological and Hydrological Institute (SMHI) on 15 November 2023, for the observed fragmented flow regime data which covered the majority of rivers throughout the country. Gauging station sites were selected on the basis of data consistency; sites were required to have no more than 5% of missing values across their entire time series, have at least ten years of data and still be collecting data on the date of data extraction. For the selected sites, we also downloaded modelled free-flowing regime data (generated by the S-HYPE hydrological model developed by the SMHI) for matching stations that were constrained between 2010 and the present day^{27,28}. The discharge data were retrieved using a custom API (application programming interface) connection script to access the SMHI open data portal (<https://opendata.smhi.se>), whereas the free-flowing data were retrieved from the SMHI REST (representational state transfer) service²⁹. We used the R programming language, v4.3.2, in all steps of the data processing and analyses³⁰.

Delineation of flow regimes to assign river ecosystem concepts

Hydrological gauging stations often provide high-resolution flow data across multiple years that can reliably relate discharge to flood inundation. Applying the River Wave to use an entire time series of data requires one key set of assumptions. The relation between discharge and associated river stage to identify the appropriate river conceptual model as outlined by Humphries et al.¹⁵ must come from the probabilistic assumptions associated with flood-frequency analysis (that is, the flood pulse concept, river continuum concept and riverine productivity model are all related to a probability distribution that determines the magnitude of a flood event). More specifically, this means that the recurrence interval of varying flood magnitudes corresponds to a relative stage height and corresponding interaction with the landscape.

To delineate the flood pulse concept (highest discharges with floodplain interaction), the river continuum concept (moderate discharges with riparian interaction) and the riverine productivity model (baseflow discharges with ordinary high water mark interaction), two discharge breakpoints are needed. We used theoretical breakpoints calculated from flood-frequency analysis to define the conditions where the flood pulse concept is most relevant; that is, any discharge greater than the mean annual flood for an entire time series. According to the theory of largest values, a mean annual flood is equivalent to the 2.33-year recurrence interval³¹. This theoretical value assumes that in a 1,000-year period, a perfect distribution for flood events can occur; however, for our study area with dams and numerous other anthropogenic impacts that reduce flood magnitude, we assume a conservative value of 2. The breakpoint between the riverine productivity model and the river continuum concept is assumed to be the median discharge across the entire time series, which can be approximated with a one-year recurrence interval. The statistical procedure to accomplish this was carried out using a Bayesian flood-frequency approach³², which fitted a log-normal distribution to flood events³³ (Supplementary Information, section 1). These assignments were used to visualize the River Wave for objective (1) and summarized to provide the proportions for objective (2).

River ecosystem concept changes across time

To compare the importance of the three ecosystem concepts for observed fragmented rivers and modelled free-flowing rivers over a full year, we modelled the ecosystem concept proportions using linear regression. Regression proportions for the observed fragmented data and the modelled free-flowing data were grouped by day of year and averaged for each river ecosystem theory for every gauge station. The corresponding day-of-year values were then fitted with a linear model where the free-flowing proportion was the response variable to the fragmented proportion. A 1:1 line was added to the graphs to show how each river ecosystem theory was biased towards either fragmented or free-flow conditions. Finally, a colour gradient was applied to the data points to visually show the seasonal patterns within a calendar year. Model summary statistics for each river ecosystem concept are provided in Supplementary Information, section 2.

River Wave metrics for organism patterns across space

We examined the River Wave metrics using hydrological methods that describe river flow regimes for both datasets in the context of riverine fish life history strategy^{34–36}. The choice to use fish life history as the focus of River Wave metrics is to remain consistent with past river ecosystem concepts, which also prioritized them as focal organisms for describing ecosystem processes and important flow events. Our implementation of the framework is not limited to fish, and similarly the metrics that we used are not the only ones available from the River Wave. For our case study, we prioritized aligning the River Wave metrics with Sweden's fish communities on the basis of an adapted version of

Winemiller and Rose's original life history strategies¹⁷: (1) equilibrium: high juvenile survivorship that is associated with stable flow conditions; (2) periodic: high fecundity that is associated with regular flow events and timing; and (3) opportunistic: short generation time that is associated with highly disturbed environments and varying flow regimes³⁷.

The River Wave metrics of baseflow consistency, peak flood timing and flashiness are routinely used to link hydrological patterns to the life history trade-offs in other studies^{38–40}. We assumed that the River Wave metrics can serve as appropriate proxies for life history strategy definitions (Table 1)⁴¹. We used the skewness of a river's baseflow index to measure the consistency of flow regimes (that is, as a proxy for equilibrium fish). We used the coefficient of variation of the annual peak flooding date across a station's time series to measure the peak flood timing (that is, as a proxy for periodic fish). Finally, we used the average Richards–Baker flashiness index value to measure volatility³⁴ (that is, a proxy for opportunistic fish). Details on life history strategy rationale and implementing these metrics can be found in Supplementary Information, section 3. As the River Wave metrics are related but typically exhibit a commensurate relationship with one another (for example, a river system with consistent large floods is less likely to have a flashy hydrograph and stable baseflow throughout the same time period), it presents an opportunity to holistically present the River Wave metrics using a ternary plot. This plot enables the proportional relations between each of the metrics to be presented in an analogous way in which the Winemiller and Rose triangle presented the original life history strategies.

Joint analysis of concepts and metrics to inform management

To utilize both the shape and area of the individual River Wave visualizations, we joined their respective data from objectives (2) and (3) to provide a joint understanding of river ecosystem. This joint analysis could be done in a variety of ways to explore management priorities. For our study we chose to identify different possible river management regimes across the country, by taking the previous calculations on life history strategies, river ecosystem model proportions and catchment size (from the gauge station metadata) to group similar flow regimes using cluster analysis. All variables were scaled before clustering to remove biases associated with larger values. We used a hierarchical clustering approach with a Euclidean distance measure and a Ward2 clustering algorithm. Clustering was done with 100 bootstrap samples with a maximum of ten clusters possible. Gap statistics and visual inspection of clusters informed the optimal number of clusters. To validate the identified clusters, we used a *k*-means clustering algorithm that iteratively applied 1–10 clusters per analysis with each output being assessed with a gap statistic. Importance of variables was assessed using principal component analysis. Details on implementation and diagnostic plots are provided in Supplementary Information, section 4.

Data availability

The observed gauge data and modelled free-flow data used in this Article are available via figshare at <https://doi.org/10.6084/m9.figshare.33087509> (ref. 42).

Code availability

Associated R functions are available via GitHub at <https://github.com/henryhansen/Riverwave>.

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Author contributions

All authors helped to conceive the study design approach. H.H.H. and J.E. led the analysis and programming. M.L. advised and supervised the analysis and methods design. L.M. and P.H. advised and supervised the theoretical aspects of the River Wave Concept. E.B. advised and supervised the case study implementation. All authors contributed to the writing and editing of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Max Lindmark.

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